Probabilistic Question Classification

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Abstract

The main purpose of question answering (QA) is to find an accurate answer to a given question. In order to correctly answer a given question it is necessary understand to a certain degree what type of information the question is asking for; that is to say, what type of question is being asked. This information can be obtained through a good question classification. This stage is very important in QA since it imposes constraints on the searched-for answer. In this paper we propose a question classification based on Markov Model that in spite of its simplicity, shows promising results.

1. Introduction

QA has become a powerful paradigm in Artificial Intelligence (AI) extending beyond AI systems to query processing in database systems and many analytical tasks that involve gathering, correlating and analyzing information. In addition, with a continuously growing explosion of information available on the world-wide web (WWW), QA is a compelling framework for finding information that closely matches the user's needs. In order to successfully match those needs, a QA has to 'understand' what the user is looking for.

As an endeavor to better understand what information the user is requiring, we have proposed 'chattering functions' applied to QA [1]. Previous research on QA [2], has shown the importance of a good question classification with respect to its answer type since it leads to the narrowing of the expectrum of candidate answers.

Question classification based on learning algorithms like Sparse Network of Winnows (SNoW) [2] and support vector machine (SVM) [3] has reported good results when classifying a question. However, computationally expensive syntactic and semantic analyses and hard labor creation of linguistic knowledge are huge. A different approach using language modeling [4] has been done but the sparseness problem was out of the scope of that report.

N-grams have been applied to the text classification task obtaining good results, therefore we propose a probabilistic question classification method based on question's valuable feature extraction and Markov assumption in order to classify a given user question. We obtained comparable results with minimal linguistic analysis.

2. Methodology

We define question classification as the task that, given a question, selects from *n* clusters the one in which that question is more probable to appear. Those *n clusters* represent *n categories*.

Let $C_a = w_1^m = w_1 w_2 w_3 ... w_m$ be a cluster '*a*', where w_i *(i=1,2..m)* are the words in that cluster, we built a first-order and a second-order Markov Model for each cluster applying the chain rule of probability:

$$P(C_a) = P(w_1)P(w_2|w_1)P(w_3|w_1^2)...P(w_n|w_1^{n-1})$$
$$= \prod^n{}_{k=1} P(w_k|w_1^{k-1})$$

where n = order of the Markov Model + 1.

In order to deal with the problem of unseen or unknown words, we used a combination of Backoff with Good-Turing discounting technique. Such as for calculating a tri-gram probability inside the cluster 'a', we have:

$$\boldsymbol{P}(w_i|w_{i-2}w_{i-1}) = \begin{cases} \wp(w_i|w_{i-2}w_{i-1}) & \text{if } T(w_{i-2}w_{i-1}w_i) > 0 \\ \propto(^{n-1}{}_{n-2})\ \wp(w_i|w_{i-1}) & \text{if } T(wi\text{-}2wi\text{-}1wi)<0 \text{ and } B(w_i|w_{i-1}) >0 \\ \propto(w_{n-1})\ \wp(w_1) & \text{otherwise} \end{cases}$$

where $T(w_{i-2}w_{i-1}w_i)$ = *generic tri-gram count.*

$B(w_i|w_{i-1})$ = *generic bi-gram count.*

$\wp(w_n|w_{n-N+1}^{n-1}) = Z^*(w_{n-N+1}^n) / Z(w_1^{n-N+1})$ is the discounted conditional probability. $Z^* = (r+1)\ M_{r+1} / M_r$ is the discounted account of a specific n-gram. M_r is the number of n-grams that occur *r* times.

$$\propto(w_{n-N+1}^{n-1}) = 1 - \sum w_n{:}m_{(wn-N+1}{}^n{}_{)>0} \wp(w_n|w_{n-N+1}^{n-1}) \ / \ 1 - \sum w_n{:}m_{(wn-N+1}{}^n{}_{)>0} \wp(w_n|w_{n-N+2}^{n-1})$$

is the probability mass to distribute from an n-gram to an (n-1)-gram. Then we calculate the discount coefficient (weight taken from already seen n-grams to be assigned to the probability of unseen or unknown n-grams):

$$D = Z^* \ / \ r$$

Since the Markov Model suffers from sparseness (Zipf's Law), a given question is pre-analyzed in order to extract "valuable features" from it. Those features are named entities, nouns and adjectives. For each cluster a frequency account of features are created. When a new question comes, its features are extracted and the probability of those features to appear in each cluster is ranked. Then the clusters where the frequency of those features is high are selected. Finally the probability of the new question to be generated by the selected cluster is calculated for the first-order and second-order Markov Model. We interpolate the obtained probabilities through a linear combination of the models such as the final probability of a question 'K' to belong to a cluster 'C_a' could be obtained as: $P(K|C_a) = \lambda_1 P_S(K|C_a) + \lambda_2 P_F(K|C_a)$ where $P_S(K|C_a)$ is the second-order Markov Model probability and $P_F(K|C_a)$ is the first-order Markov Model probability. The weights λ_1 and λ_2 vary depending on the cluster. In general $\sum_{i=1}^{2} \lambda_i = 1$.

3. Experiment and Results

As training data, 3865 questions from a corpus publicly available [2] were selected. Those questions were distributed into our 24 clusters in order to build the first-order and second-order Markov Model. As test data, 250 TREC 10 questions were distributed into 24 sets according to each cluster. With this experiment we tried to determine how well each set of question was generated by the clusters, that is to say, the probability of each set to belong to a specific cluster. Results showed that with this methodology 21 of 24 sets where correctly classified according to its cluster or category which means 91.6% of accuracy for the classification of the sets. Looking inside each set the accuracy of classifying individual questions was found to be 81.3%, which is comparable with results obtained from other systems [2], [3], [4].

4. Conclusion

We propose a probabilistic method for question classification and in spite of the simplicity of the system, results showed an acceptable performance. Further improvements are oriented to a better question analysis including "word focus" identification, in order to lead to a better classification accuracy.

References

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